

UAV Fleet Area Coverage with Network Connectivity Constraint

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ABSTRACT

An Unmanned Aerial Vehicle (UAV) is an aircraft without onboard human pilot, which motion can be remotely and / or autonomously controlled. Using multiple UAVs, i.e. a fleet, offers various advantages compared to the single UAV scenario, such as longer mission duration, bigger mission area or the load balancing of the mission payload. For collaboration purposes, it is assumed that the UAVs are equipped with ad hoc communication capabilities and thus form a special case of mobile ad hoc networks. However, the coordination of one or more fleets of UAVs, in order to fulfill collaborative missions, raises multiples issues in particular when UAVs are required to act in an autonomous fashion. Thus, we propose a decentralised and localised algorithm to control the mobility of the UAVs. This algorithm is designed to perform surveillance missions with network connectivity constraints, which are required in most practical use cases for security purposes as any UAV should be able to be contacted at any moment in case of an emergency. The connectivity is maintained via a tree-based overlay network, which root is the base station of the mission, and created by predicting the future positions of one-hop neighbours. This algorithm is compared to the state of the art contributions by introducing new quality metrics to quantify different aspect of the area coverage process (speed, exhaustivity and fairness). Numerical results obtained via simulations show that the maintenance of the connectivity has a slight negative impact on the coverage performances while the connectivity performances are significantly better.

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Mobility models, UAV networks, Quality metrics

1. INTRODUCTION

An Unmanned Aerial Vehicle (UAVs), also referred to as drone, is an aircraft without a human pilot onboard. Its flight is either controlled autonomously by computers in the vehicle, or under the remote control of a pilot on the ground or in another vehicle. The development and usage of UAVs quickly increased in the last decades, mainly for military purposes. Nowadays, this type of technology is already in place in civil contexts, essentially for civil and environment protection: search & rescue teams, fire fighters, police officers, environmental scientists, etc. Although the technology for operating a single UAV is now mature, efforts are still necessary for using UAVs in fleet.

Using multiple UAVs offer various advantages compared to the single UAV scenario. First, a fleet can theoretically propose unlimited mission duration, i.e. a subset of the UAVs are refilling their energy reserves (fuel, electricity), while the rest of the fleet continues fulfilling its objectives. Cooperative fleets of UAVs also help to increase the maximum size of the mission area, as nodes can relay communications in an ad hoc fashion, when they are out of the base station's transmission range. Moreover, using fleets allows balancing the total payload among the different UAVs, for some specific missions. Finally, a fleet permits to fulfil missions that cannot be satisfyingly performed with a single UAV, e.g. when simultaneous measurements are necessary, or when an area is too wide to be covered by a unique UAV frequently enough.

Most of the use cases for fleets of UAVs are related to coverage problems, i.e. the fleet should deploy itself and scan a designated area to detect the presence of something, such as illegal activities, lost people, etc. As a consequence, the success of a mission highly depends on the quality of the deployment, i.e. the way the UAVs are scanning the area. In the literature, some researches have been conducted to design trajectories for groups of UAVs. Most of the current approaches are considering both offline and online centralised computations. However, we think that excepted for very specific missions, this is not recommended. Indeed, neither unexpected technical issues nor the weather conditions or even the attack of an aircraft by hostiles can be predicted precisely enough and thus, offline approaches cannot be applied to most of the use cases. Online mission planning is highly suitable for coverage missions, however, continuously maintaining the flight plan of a fleet via a unique access point, e.g. satellite connection, may also create problems in terms of robustness or communication costs. Indeed, what would happen if the central entity fails or is not able to communicate with the fleet frequently enough? For all these reasons, we focus on **online and decentralised techniques**.

To this end, every aircraft is assumed to be equipped with a wireless transceiver to communicate with its neighbours, i.e. the aircrafts that are currently in the transmission range of the transceiver. This new type of **ad hoc network** is thus composed of a set of mobile UAVs and a fixed base station, which acts as the surveillance headquarter. Each UAV will thus be able to exchange information on a peer-to-peer fashion and determine its flight plan solely based on its aggregated local knowledge. Gathering a complete knowledge of the system would decrease the scalability of the approach, as in such networks, this would require a continuous broadcasting process of all the aircrafts to maintain everyone updated and thus would dramatically reduce the available bandwidth for application-related communication. This decentralised approach is characterised by its robustness as the surveillance process continues in spite of potential technical failures or destruction.

Although online and decentralised approaches have been proposed, the domain remain quite unexplored, in particular concerning the comparison's methodology. Indeed, very few quality aspects of these approaches have been formalised with quality measures, thus leading to un-compared contributions. In order to assess this problem, we propose a set of new quality metrics designed to quantify various aspect of the coverage process, i.e. speed, exhaustivity and fairness.

Although the usage of UAVs and fleet of UAVs is expected to increase exponentially in a near future, many regulatory issues are still being worked on. Currently, most safety-related regulations require that any UAV should be able to be controlled at any moment in case of an emergency. It induces that the UAV's flying range should be limited to its transmission range. One of the advantage of multi-UAVs scenarios is however to expand the range of the missions. In this context, it is thus required to maintain a connected network to ensure that the remotest UAVs can still be controlled via multi-hop communications. To the best of our knowledge, no mobility models for such networks with connectivity constraints has been published so far. In this work, we propose a new mobility model to efficiently cover a designated area while maintaining a connected network in a decentralised and localised fashion. We then validate its

performances according to the aforementioned quality metrics and we compare it to relevant mobility models from the literature.

The remainder of this paper is organised as follows. In Section 2 we review the relevant literature. Then, Section 3 presents the area coverage problem with connectivity constraints and its related optimisation objectives. Quality metrics for this specific problem are then formalised and discussed in Section 4. The connected coverage mobility model is described and explained in details in Section 5 and its experimental validation is then exposed throughout Sections 6 and 7. Finally, we conclude this paper and discuss some future re- search directions in the last section.

2. RELATED WORK

In this work we assume that all the UAVs are equipped with transceiver allowing inter-UAV communication in a peer to peer fashion. As a consequence, we consider that a fleet of UAV is a subclass of mobile ad hoc and sensor networks. Many practical use cases of ad hoc networks, including the one considered in this work, are characterised by the mobility of their network components. As a consequence, many research works have been initiated to deal with, or to leverage issues induced by the mobility of the network nodes. The mobility was assumed to be unpredictable and uncontrollable, thus, these works, which generally were extending or adapting methods and techniques designed for static ad hoc networks, considered mobility as a curse. As the motion of the nodes in these works is assumed not to be controllable, some techniques have been developed to try to predict the mobility in order to take more sustainable decisions through time [1, 9, 12]. These methods generally named themselves mobility aware or mobility management mechanism [4, 12]. However, even when mobility could be predicted, it was assumed that its influence was degrading the quality of service in the network. More lately, the mobility was rehabilitated by Basagni *et al.*, who stated explicitly that mobility should be considered a blessing for ad hoc networks in [2]. The authors argued in favor of mobility as various works proved the inherent benefits of mobility in ad hoc networks, such as the increase of the data carrying capacity, the increase of the network lifetime, a better data fidelity or even a better resilience to malicious attacks. More recently, there has been an increasing interest in developing techniques and algorithms related to the usage or control of the mobility in ad hoc networks in general, and in sensor networks in particular. As explained previously, mobility can be envisaged as a blessing considering some induced networking properties such as network capacity or lifetime. It is fair to assume that if uncontrolled mobility helps, controlled mobility may be even more beneficial. These contributions will be referred as Mobility Control (MC) techniques.

In [5], Ekici *et al* proposes to classify mobility for wireless sensor networks depending on what entity is actually moving. Three categories are introduced: mobile base station (MBS)-based solutions (e.g. [6]), mobile data collector (MDC)-based solutions (e.g. [11] and rendezvous-based solutions [7]. In these contributions, the mobile elements are generally a single sink and the difference lies in how the sensor data are aggregated. In our work, the scenario is the completely opposite situation: all the entities of the network are moving except the sink, here the base station.

Some works have been conducted about path planning for multiple UAVs (see [3]). However these works are intrinsically offline approaches which we think are not suitable in such a dynamic environment. The closest work we found so far is from Kuiper *et al.* [8], in which they designed decentralised algorithms in charge of moving UAVs for coverage missions. Two approaches are briefly compared, a random choice between three directions (left, front, right) and a more sophisticated one using pheromones in order to attract UAVs where they are the most needed. This work is very interesting but does not consider the maintenance of the connectivity which is important for our scenario.

3. PROBLEM DESCRIPTION

In this work we tackle a classical use case of UAV, namely surveillance. We have an area that needs to be monitored and a fleet of UAVs in charge of patrolling it. This fleet is deployed from a base station and all UAVs are equipped with communication capabilities (with the base station and the other UAVs). Our focus is to design mobility models, i.e., algorithms in charge of physically moving the UAVs in order to fulfill their common mission. This mission is characterised by two main objectives: covering the area and maintaining the connectivity in the communication network formed by the UAV. The later objective is particularly important for safety reasons as it is generally required by air regulations to be able at any moment to bring back a flying entity to the base station or at least to land it safely. Thus, we will focus on the following four objectives to assess the quality of the proposed mobility models:

- The speed of the coverage: the ability to cover an area as fast as possible.
- The exhaustivity of the coverage: covering the totality of the surveilled area.
- The fairness of the coverage: how even is the covering of the area.
- The maintenance of the connectivity.

4. QUALITY METRICS

In this section, we propose metrics to assess the quality of the mobility models with regards to the aforementioned objectives. We think that this is mandatory, since the mobile behaviours of the literature are generally validated based on project-specific measures and not compared with other contributions on their raw performances with standardised metrics. As a consequence, this section proposes different quality measures encompassing various important criteria of a typical area coverage mission, such as the speed, the exhaustivity and the fairness of the coverage process. Additionally, we introduce metrics for the connectivity preservation aspect which is the specificity of our approach.

4.1 Definitions

Let $T = t_0, \dots, t_{max}$ be the set of all simulation time-steps. For the sake of simplicity, the considered simulation area is assumed to be a rectangle named A , with a long side L_{long} and a shorter side L_{short} . We discretise the area into square-shaped cells of side L_{cell} such as $L_{long} \cong L_{cell}$ and $L_{short} \cong L_{cell}$. Let $nbLS = \frac{L_{long}}{L_{cell}}$, respectively $nbSS = \frac{L_{short}}{L_{cell}}$, be the number of cells fitting in the long side, resp. short side, of the area. Let C be the set of all the cells,

such as $C = \{c_{i,j}\}$, with $1 \leq i \leq nbLS$ and $1 \leq j \leq nbSS$. Let $nbCells = nbLS * nbSS$ be the total number of cells. The size of the cells' side impacts on the granularity of the simulation. Indeed, the bigger is L_{cell} , the less cells will fit into the simulation area, and thus, the simulation results will be less precise. Let $Scan : C \rightarrow T$ be the function that returns the last time at which a cell has been scanned by a UAV. $Scan$ returns -1 if a cell has never been scanned yet. A time-dependent version of this function, $Scan : C, T \rightarrow T$, returns the last time at which a cell has been scanned by a UAV in the time interval $[0, t \in T]$. Let $nbScan : C \rightarrow \mathbb{N}$ be the function returning the number of times that a given cell has been scanned.

4.2 Speed of the coverage

It is generally assumed that a fleet should cover the surveilled area as quick as possible, i.e., it is highly suitable to reduce the time necessary to scan all the cells at least once. The following metric allows to compare different mobile behaviours concerning their ability to quickly perform a first complete scan of the simulation area from the deployment time t_0 .

This first metric can be formally defined as:

$$speed = \min t \in T / \forall c_{i,j} \in C, Scan(c_{i,j}, t) \geq 0. \quad (1)$$

Note that the strict constraint of this metric, i.e. the entire set of cells, can be relaxed by considering a predefined percentage of cells of the total area. The following metric proposes a way to restrain the coverage to a given percentage p :

$$speed(p) = \min t \in T / \frac{|\{c_{i,j} \in C, Scan(c_{i,j}, t) \geq 0\}|}{|C|} > p. \quad (2)$$

4.3 Exhaustivity of the coverage

The exhaustivity of the coverage refers to the amount of area that has never been scanned during a simulation run. This metric can be seen as the surface-oriented version of the speed metric which is time-oriented. Indeed, while speed is considering the required amount of time to scan a given percentage of the area, the exhaustivity measures how many cells have never been scanned.

This metric was proposed during the experimental phase of this work in order to obtain information when we discovered that the speed metric was not returning any value when the total area was considered. Instead, we considered speed for 90% and 95% of the area and we added the exhaustivity metric to get a more precise idea of the mobile behaviour results. The exhaustivity metric can be formalised as follow:

$$ex = \frac{|\{c_{i,j} \in C, Scan(c_{i,j}, t_{max}) = -1\}|}{nbCells} \quad (3)$$

4.4 Fairness of the coverage

An additional important issue for scanning an area which is not taken into account in the former metrics is the fairness of the coverage, i.e. are all the cells regularly and equally scanned? Indeed, it is completely possible to obtain good results with the previous metrics while having cells which are almost never scanned and some others intensively scanned. In order to keep track of that we introduce the notion of fairness of the coverage and we define it as the dispersion of the number of scans for all the cells of the area. This can be

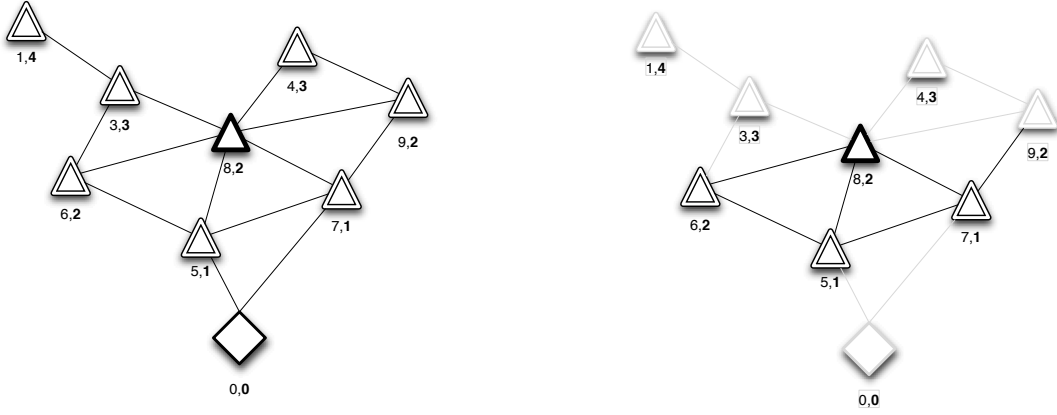


Figure 1: Selection of the considered neighborhood.

computed by the standard deviation, for all the cells, of their respective number of scans. The metric is thus formalised as follow:

$$fair = \sqrt{\sum_{c \in C} \frac{(\overline{nbScan} - nbScan_c)^2}{|C|}} \quad (4)$$

Obviously, the distribution of the number of scans among the cells does not take into account the fact that cells may have been visited intensively at the beginning of the simulation and more rarely later. However, from our point of view, characterising such behaviour would require to compute one value per cell and all this information could disappear during the aggregation process to a unique value for the whole map. Although we have not found the perfect solution yet, we continue to investigate.

4.5 Connectivity

An important issue when dealing with flying object is the security, i.e., being able to switch off at any time and in a safe manner any flying components. Although safe landing is not in the focus of this work, being able to contact a particular UAV at any moment to order an emergency landing can become handy. Indeed, in the type of network we consider, it is frequent for a UAV to be out of the base stations' transmission range so that remote places can be scanned in the surveyed area. As a consequence, some other UAVs should relay potential transmissions to and from the base station and thus a communication path should be available at any moment.

The network we are considering can be modeled as a dynamic graph $G_t = (V_t, E_t)$, in which the vertices $v \in V_t$ represent the actors of the network, i.e. the multiple UAVs and the base station, and the edges $e \in E_t$ represent potential wireless communication links. The dynamicity implies that the vertex and the edge set can vary through time. Such a graph is generally referred as **communication graph**. If a communication path exists between any pair of vertices in the graph, then it is said to be connected. This can be verified easily by any graph search algorithm in a computationally efficient manner $O(n \log m)$, with $V_t = n$ and $E_t = m$.

In this work, we propose two connectivity-related metrics: the number of connected components and the percentage of

nodes connected to the base. Both metrics can be computed for any $t \in T$ and for the sake of analysis, average and standard deviation values will be extracted.

Number of connected components.

A connected component of a graph is any maximal set of vertices which are pairwise connected by a path. Computing the number of connected components helps us to obtain information about how much disconnected is a graph. The optimal (and minimal) value for this metric is thus 1. Let $nbCC_t$ be the number of connected component at a time $t \in T$.

Percentage of UAVs connected to the base.

However, the same number of connected components may hide very different situations. For example, there may be two connected components but in the first case 95% of the UAVs are connected with the base station and in another one only 5%. In order to detect such variations, we propose to keep track of the number of UAVs connected (directly or indirectly) to the base station. Let $percBS_t$ be this percentage.

5. CONNECTED COVERAGE MODEL

In this section we present in details the main contribution of this work, i.e. the connected coverage mobility model. This model is composed of three sequential steps: a selection of the considered neighborhood with which the UAV will try to stay connected (1), the computation of the viable alternatives (directions) concerning the future position of the UAV (2) and a pheromone-based behaviour to select the best direction from the previously computed set (3).

5.1 Neighborhood selection

All UAVs are characterised by their graph distance to the base station, their current position, heading and speed, and a unique identifier. The proposed connected coverage mobility model is maintaining a tree structure which root is the base station used in our scenario. Each UAV is characterised by a couple (id, hop) composed of a unique identifier (id) and the number of hops to the base station (hop). A UAV u is inferior to another UAV v , $u \prec v$ if $hop_u < hop_v$ or $(hop_u = hop_v) \wedge (id_u < id_v)$.

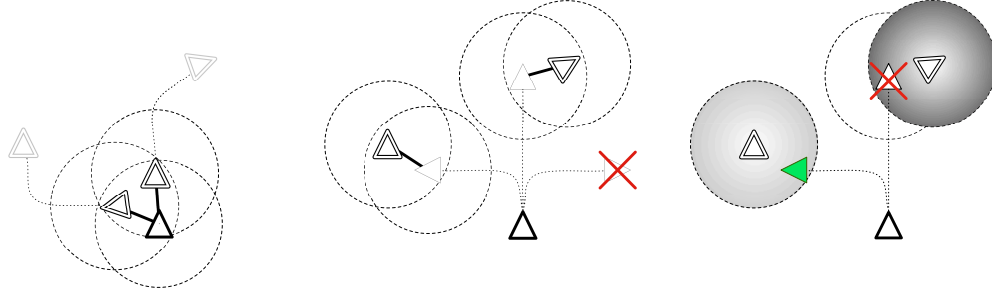


Figure 2: The computation of the viable alternatives and the selection of the best one.

The basic idea is that each UAV u considers a specific neighborhood $N_{cc}(u)$ composed of its one-hop inferior neighbours to select a future destination for which u would be connected to at least one UAV from $N_{cc}(u)$. In Figure 1, the base station is represented by a square while the UAVs are triangles. The considered UAV is here represented in bold. Its identifier is 8 and it is two hops away from the base station. On the left side of the Figure we can see the complete communication graph at a given instant. The right side of the Figure shows the UAVs which will be considered for the next steps (computing a viable alternative and choosing the most efficient one). Here, we can see that not all its one-hop neighbours are considered. Indeed, only inferior one-hop neighbours are selected. The selection is first done of the number of hops to the base and in case of equality, a tie-break rule uses the unique identifiers.

This mobility model works in a decentralised manner with local (partial) knowledge. For this purpose, each UAV periodically takes decision about his next destination based on informations from its one-hop neighbours, i.e.

5.2 Computing viable alternatives

This step is in charge of computing the future potential destinations that a considered UAV can choose in order to remain connected to the UAV network. These destinations are computed thanks to the following steps. First, from the current position and the current destination of every one-hop neighbour n_i , we estimate where n_i will be in a predefined future. Then, we check where we would be in this predefined future depending on what direction we choose. In Figure 2, a very simple example of the process is illustrated. The considered UAV is again represented in bold while its neighbors are drawn with double lines. The transmission range is represented by a dashed circle. In this work we consider the simplistic situation where the transmission range is a perfect circle and the transmission power is uniform for all UAVs. As a consequence, a communication channel is always bi-directional. In this scenario, two UAVs are one-hop neighbours if and only if they are in each others' transmission range. On the left side of the Figure, the present moment is represented. The considered UAV knows where its neighbours currently are along with their respective current destination. Thus, it can compute their potential position in the considered future. Here, these future positions are represented in light grey. The middle section of the Figure represents the actual computation of the viable alternative. In our simple example, only three directions are evaluated (left, front and right). Note that in our implementation,

we increase this number to one hundred in order to obtain a diversified set of alternatives. The considered UAV computes its future position according to these three directions, here drawn with a thin black line. Only two directions are proposing viable alternatives (left and front), right is discarded for the next and last step.

5.3 Selecting the best alternative

When we have a set of viable alternatives, we need to decide which one is the best considering the operational objectives, i.e. the surveillance of a designated area. We decided to use the mechanism introduced in [8] which is using an Ant Colony behaviour in order to guide the UAVs in their decision. We chose this approach for its simplicity and its efficiency. The objective is here to attract UAVs in parts of the area that have not been scanned recently thanks to pheromones, a chemical information varying in intensity. In our scenario, UAVs will regularly drop pheromones on their way and these pheromones will gradually disappear through time. The UAVs will thus be attracted by places that are characterised by the lowest pheromone concentration. In this situation, as the terrain cannot be used for storing pheromones, the UAVs are required to share their local knowledge of the scanned area via periodical messages. In Figure 2, the right side represents this last step. We can see that the left alternative is better as the pheromone concentration, here represented in light grey, is less strong than for the front alternative, here in dark grey.

6. EXPERIMENTAL SETUP

In this section, we first present the simulation environment along with the most important simulation parameters. In a second time, we briefly present the two mobility models that we use in our comparison, i.e. a random movement and a pheromone-based mobility model.

6.1 Simulation environment

The simulation environment is fully based on a dynamic graph library named Graphstream [10]. In this subsection we provide details concerning the implementation of the area, the autopilot of the UAVs and the main simulation parameters.

6.1.1 The Area

As previously said in subsection 4.1, the simulation area is a rectangle with a long side L_{long} and a shorter side L_{short} . It is discretised into square-shaped cells of side L_{cell} such as $L_{long} \cong L_{cell}$ and $L_{short} \cong L_{cell}$. The size of the cells' side

impacts on the granularity of the simulation. Indeed, the bigger is L_{cell} , the less cells will fit into the simulation area, and thus, the simulation results will be less precise. Table 1 summarises the values of the parameters we used during our experimentations.

6.1.2 Autopilot

The implemented autopilot mimics a simple fixed-wing aircraft behaviour and thus is characterised by minimum and maximum speed limits and also a maximum turning rate. It only requires the setting of a destination to operate, i.e. automatically set the current speed and heading of the aircraft. When the UAV is far from the destination, the autopilot progressively modifies the heading until being aligned with it. Then the UAV will proceed in straight line. When the UAV arrives at destination, if no new destination is set, the aircraft will fly in circles as a fixed-wing aircraft cannot hover. The mobility model in charge of modifying the current destination is called every 30 time-steps in order to limit the computer power requirements. This value has been chosen after an extensive set of experiments in which a wide range of values have been tested. The main parameters related to the flight behaviour of the UAVs are show in Table 1.

6.1.3 Simulation parameters

As all the mobility models are stochastic, we repeated each set of experiments 30 times in order to obtain a good statistical confidence. For all quality metrics we then aggregated the average and the standard deviation. In this work, a set of experiment is a couple (mobility model, number of UAVs). As it can be seen in Table 1, we conducted experiments with very different density conditions as we start at 10 UAVs up to 50 with a wireless transmission range fixed at 400.

6.1.4 Limitations

The aforementioned experimental setup is based on a custom-made simulator based on a dynamic graph library. As a consequence, it should be reminded that the physical models for fixed-wing UAVs cannot be considered as highly realistic from an aerodynamic point of view. Additionally, the wireless communications are assumed to be idealistic, i.e. no collision or interferences and unlimited bandwidth, as in [8]. Combining a flight simulator with a wireless network simulator certainly would have help obtaining more precise results but we think that the realism level is sufficient for assessing the performances of the tested mobility models.

6.2 Compared Mobility Models

We compare our mobility model with the two proposed in [8], i.e. a random mobility model and a pheromone-based mobility model. This subsection briefly introduces their main characteristics.

6.2.1 Random Mobility Model

The random mobility model considers three potential directions on a regular basis: front, left and right. For each direction, a destination point is computed by using the autopilot software, i.e. this point can actually be physically attained by the UAV regarding its aerodynamic constraints. Depending on fixed probabilities, the model then decides which point will be chosen as the current objective destina-

Parameter name	Parameter value
Simulation area	
L_{long}	2000
L_{short}	1000
L_{cell}	20
Base station position	(1000,0)
UAV Autopilot	
Speed	[0 .. 10]
Acceleration	[-1 .. 1]
Heading Change	[0 .. 0.1]
Decision Frequency	30
Future	30
Wireless range	400
Experiments	
Number of UAVs	[10 .. 50]
Number of runs per experiment	30

Table 1: Main simulation parameters

tion point set in the autopilot. Like in [8], we set a higher probability for the front (60%) than for both turning directions (20%).

6.2.2 Pheromone-based Mobility Model

The pheromone-based mobility models use the idea of stigmergy, i.e. the ability to coordinate agents in an indirect fashion based on the exchange of information (pheromones) in the environment. In our case, the environment, i.e. the area to be scanned, can not be used for exchanging pheromones.

As a result, we rely on a discrete representation of the territory via a map, i.e. the area is broken down to small sub-parts and each of them can be characterised by a pheromone level. On the implementation level, this model is quite similar to the random mobility model as a decision is taken between three different destination points: front, left and right. Here however, the UAV chooses the destination that is characterised by a lower pheromone level. In case of tie, the final decision is made randomly between the destination with equal pheromone levels.

7. NUMERICAL RESULTS

In this section we summarise the obtained numerical results for our mobility model and we compare its performances with both the random and the pheromone-based mobility introduced in [8].

7.1 Speed of coverage

Not surprisingly, the random mobility model does not perform well considering the speed of the network, whatever the size of the simulation area or the number of UAVs in the simulation. Figure 3 illustrates these results. Our mobility model performs very well for the coverage speed even though the results are less good when there are very few UAVs, here 10. This bad result is mainly due to its connectivity constraint. Indeed, when they are less UAVs, the exploration process is much more difficult as less viable alternatives exist. This issue resolves itself for 20 UAVs where the performances of the pheromone-based approach and ours are the

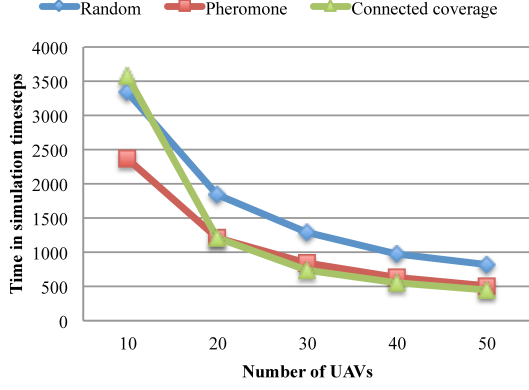


Figure 3: Speed of coverage for 95% of the area.

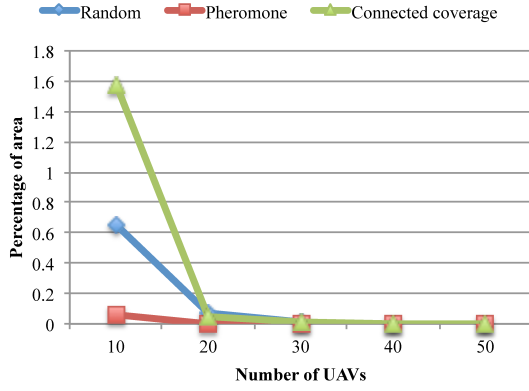


Figure 4: Exhaustivity comparison.

same. Finally, for denser networks, the connected coverage model obtains slightly better results.

7.2 Exhaustivity of coverage

For the exhaustivity metric, we can observe in Figure 4 that our mobility model does not perform well when the number of UAVs is low. As explained in the previous subsection, this is due to the very low amount of viable alternatives which drastically reduces the exploration possibilities. We can observe that the phomone-based approach works perfectly as almost 100% of the area is scanned, whatever the number of UAVs. The random mobility model, which we consider to be the lowest acceptable bound, obtains better solution than our approach for 10 UAVs and thus further investigations should be done to obtain better exhaustivity.

7.3 Fairness of coverage

In Figure 5, we can observe that that our mobility model is not behaving the same way than the others concerning the fairness of its coverage process. Indeed, it can be observed that it remains almost stable whatever the network density. When very few UAVs are in the simulation, the results are clearly not good and can be explained by the fleet being quite constrained. A few places are explored sporadically but the UAVs remain most of the time around the

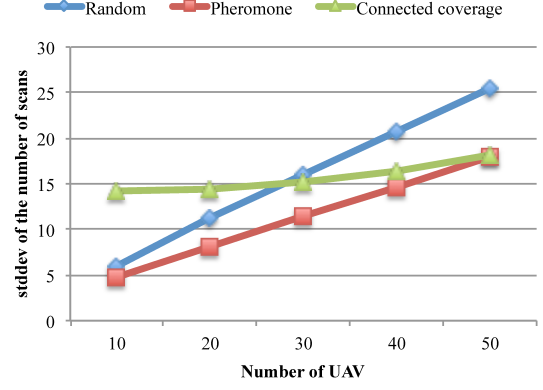


Figure 5: Fairness comparison.

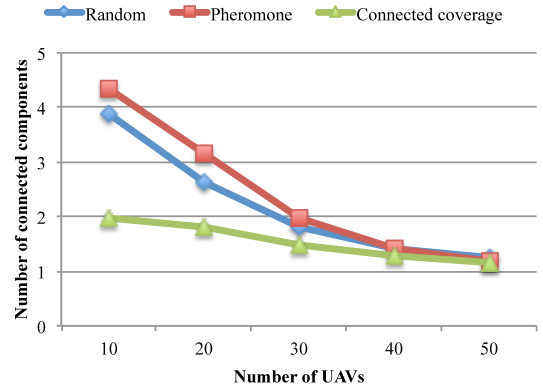


Figure 6: Number of connected components.

same positions in order to maintain connectivity. However, this behaviour remain as bad throughout the whole spectrum of density, while the others mobility models exhibit a linear progression with the number of UAVs. As a result, our approach obtains better results than the random mobility model from 30 UAVs and the results are similar to the phomone approach for the densest network. We can observe here the obvious advantage of the phomone technique over the random mobility for a more fair exploration process.

7.4 Connectivity

As can be seen on both Figures 6 and 7, our contribution clearly outperforms both compared mobility models. One may argue that the comparison is not fair for the two approaches are not targeting to maintain connectivity. However the objective of guaranteeing the connectivity in a decentralised manner is successful. First, the number of connected components always remain below 2 while the other mobility models start around 4. The density helps these connectivity-unaware models and we can observe that all of them converge for the densest networks. Second, the number of UAVs connected to the base is very high, which means that even if we have 2 connected components, we are in the case of a very big number of UAVs reachable by the base

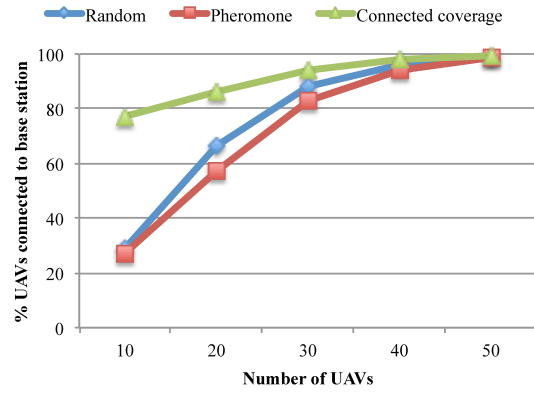


Figure 7: Percentage of UAVs connected to the base station.

station and a few temporary lost UAVs. Both connectivity-unaware models perform very poorly with respect to this last metric, in particular when the number of UAVs is low.

8. CONCLUSIONS AND PERSPECTIVES

In this work we proposed a decentralised and localised algorithm to control the mobility of a fleet of UAVs performing a surveillance mission over a designated area. In the large majority of use cases, regulations require to be able to contact any UAV at any moment in case of an emergency. To such purpose, we propose the connected coverage mobility model which maintains a tree based overlay network rooted at the base station. While maintaining the connectivity of the network, our approach also benefits from an Ant Colony Optimisation technique for the exploration process, i.e. pheromones are used to guide UAVs to the places which have been less covered. We compared our work with two mobility models from the literature by proposing new quality metrics related to the most important objectives of surveillance missions, i.e. speed, exhaustivity and fairness. Via extensive sets of simulations, we empirically proved that our mobility model was able to greatly enhance some connectivity metrics, i.e. number of connected components and the percentage of UAVs connected to the base, while still maintaining comparable performances for the others aforementioned metrics.

Although this first step can be considered as successful, many future works are envisaged. First, we noticed that the maintenance of the connectivity was way too constraining when there is a small numbers of UAVs and we think we may find less constraining rules that would allow a better exploration without impacting too much the connectivity. In this vein, we would like to adapt the behaviour's freedom depending on the distance with respect to the overlay tree structure, i.e. the more distance, the more freedom, and the number of viable alternatives. In the numerical results sections, we observed that when the density is increasing, the fairness is lowered. We visually noticed that this may be due to concurrent decisions and some delay in the collaboration process that induce multiple UAVs to choose very similar destinations. As a consequence, we are currently working on a collision avoidance mechanism that we think will help for both the fairness and the speed.

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